

A090 – Projection Chain: $T^4 \rightarrow T^0$, Three Operations

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Contents

.1	Purpose	1
.2	The Tension in the Question	1
.3	The Structure of T^4	2
.4	Type I – Duality Decomcompactification: Mass Becomes Time	2
.5	Type II – Geometric Projection: Forget Spatial Directions	3
.6	Type III – Representational Translation: Bijective Upward	3
.7	The Formal Chain, Level by Level	3
.8	Three Operations at a Glance	4
.9	The Role of the Recursion ξ^n	4
.10	Reversibility, Precisely Per Type	4
.11	Why Exactly Four Circles	4
.12	In the Compact Picture: the Time Circle Is Chosen by the Measurement	4
.13	The Topological Identification $t \sim t + R_m$ as Physical Observable	5
.14	The Standard Model as Decomcompactified Projection	5
.15	Open Questions	5
.16	Verification Scripts	6
.17	Supersedes	6
.18	Use of AI Tools	6

.1 Purpose

[STIPULATION] FFGFT describes fundamental physics on a compact 4-torus T^4 with the single parameter $\xi = 4/30000$. The observable world is $\mathbb{R}^3 \times \mathbb{R}_t$. This document addresses the question of how both are connected – and shows that the chain $T^4 \rightarrow T^0$ consists of *two qualitatively different* operations that must not be confused: a *duality decomcompactification* (nearly reversible, no mode losses) and a *geometric projection* (lossy, not reversible). A third operation, going in the opposite direction, is added: the *representational translation* to the Hilbert space (bijective, lossless).

The Standard Model is situated therein as a *decompactified projection*: its constants (α , v , Yukawa) are recursion outputs from ξ ; whoever takes them as inputs works in the unrolled picture, without naming the compact whole. This is not a refutation of the Standard Model – it computes correctly in its domain; it is a location.

.2 The Tension in the Question

[B] Three observations about the transition $T^4 \rightarrow \mathbb{R}^3 \times \mathbb{R}_t$ appear in mutual contradiction:

- **Loss picture.** Every projection $D \rightarrow D - 1$ is not injective, loses information. The chain is not reversible.

- **Compactness picture.** T^4 is fully compact; the observable spacetime is decompactified.
- **Upward picture.** There is already a *bijective* translation of T^4 into a higher-dimensional space – the Hilbert space (A070) – which is *lossless*.

[B] These three pictures cannot be naively overlaid: if the first step were pure loss, no decompactified time direction could arise from it (forgetting does not generate a new dimension); and it would remain puzzling how an upward movement can be lossless while the downward movement destroys information. The resolution: there are not one but *three* dimension-changing operations of *two kinds*.

.3 The Structure of T^4

[B] The fundamental level of FFGFT is the compact 4-torus

$$T^4 = S_x^1 \times S_y^1 \times S_z^1 \times S_m^1,$$

with three *spatial* circles and one *mass circle* S_m^1 . The fourth direction S_m^1 is not a mass dimension *beside* a separate time, but the *substitute time direction* itself. Mass and time are two *readings* of one and the same dimension; the fundamental relation $\tilde{T} \cdot m = 1$ is the translation dictionary:

$$\underbrace{S_m^1 \text{ (compact)}}_{\text{mass reading, discrete spectrum}} \xleftrightarrow{\tilde{T} \cdot m = 1} \underbrace{\mathbb{R}_t \text{ (open)}}_{\text{time reading, continuous flow}}.$$

[K] Kaluza–Klein reading. A compact circle of radius R carries a discrete mass tower $m_n \sim n/R$. The four directions carry two radius classes: the three spatial circles ($R_{x,y,z} \sim R_{\text{large}}$) generate a quasi-continuous mode tower with ~ 0 gap – this becomes space; the mass circle ($R_m \sim \xi \ell_p$) has a huge gap and a discrete, heavy spectrum – this becomes the particle mass spectrum. The fundamental relation $\tilde{T} \cdot m = 1$ is thereby structurally the Kaluza–Klein relation for the mass dimension.

.4 Type I – Duality Decompactification: Mass Becomes Time

[B] The first, qualitatively distinguished operation is the *unrolling* of the mass circle via its universal cover:

$$S_m^1 \xrightarrow{\text{cover}} \mathbb{R}_t, \quad p : \mathbb{R}_t \rightarrow S_m^1, \quad t \mapsto e^{2\pi i t / R_m},$$

governed by $\tilde{T} \cdot m = 1$. This operation is *embedding*, not projection: it generates a new direction ($\mathbb{R}_t$) instead of discarding one. Periodic structures on S_m^1 lift uniquely to $\mathbb{R}_t$; no mode content is lost. The “price” of decompactification is not information loss but the appearance of the time direction itself.

[B] What does not remain directly visible: the *compactness*. In the mass reading the topology is manifest – the spectrum is discrete, the identification $t \sim t + R_m$ directly expressed. In the time reading the same compactness is only *latent*: from continuous time alone the topological identification is not uniquely reconstructible (the winding remains ambiguous). Precisely there the recursion ξ^n sets in – it is the spectral signature by which the hidden compactness remains recognisable.

[S] What makes unrolled time continuous and uniform. Three questions, three answers: (1) *Why is $\mathbb{R}_t$ continuous?* Because the unrolling is the universal cover $p : \mathbb{R} \rightarrow S^1$ – a local homeomorphism; the cover of a circle is the continuous line. (2) *Why does time flow uniformly?* Because of the homogeneity of S_m^1 (translation invariant, no distinguished point)

and the single fixed eigenvalue $m = 1/\tilde{T}$. (3) *Why is it unrolled at all?* Not the geometry decides this – in the compact T^4 all four circles are equivalent, there is no time. The *measurement process* (the embedded observer) chooses one circle as time and unrolls it. Continuity is the forgotten winding: keeping it would give back the discrete circle.

.5 Type II – Geometric Projection: Forget Spatial Directions

[B] All further steps ($D3 \rightarrow D2 \rightarrow D1 \rightarrow D0$) are genuine geometric projections of spatial directions – a projection $\pi : \mathbb{R}^k \rightarrow \mathbb{R}^{k-1}$ with $\ker(\pi) = \mathbb{R}$. It collapses a complete fibre; the information about the forgotten direction is not contained in the image. Without additional data (holographic completeness, boundary conditions) not reconstructible. The recursion ξ^n does *not* act on these steps – spatial projection losses are genuine.

[B] The step $D3 \rightarrow D2$ is mathematically the holographic principle; in the horizon case the Bekenstein–Hawking bound $S \leq A/(4\ell_P^2)$ appears here.

.6 Type III – Representational Translation: Bijective Upward

[B] Downward (geometric) and upward (representational) are *opposite* information properties:

$$T^4 \xleftrightarrow{\text{bijective}} \mathcal{H} = L^2(T^4) \otimes \mathbb{C}^3$$

The translation (A070) is not an approximation but a bijective structural statement: every QM computation is mappable into FFGFT language and vice versa. *No* physical extra dimensions are added – the same 4D content is expressed in a higher-dimensional *language* in which it fits without reduction. Losslessness follows precisely from the representational character.

[B] A geometric extension $T^4 \rightarrow T^5 \rightarrow \dots$ is excluded in FFGFT: complexity arises from *fractal depth* ($D_f = 3 - \xi$, self-similarity), not from *more* dimensions. This is the deliberate contrast to string theory (10/11D, Calabi–Yau, landscape problem). “Higher” in FFGFT always means type III (representational), never geometric.

.7 The Formal Chain, Level by Level

[B] The chain $T^4 \rightarrow T^0$ separated by type:

Level	Operation	Type	Invariant	ξ^n
$D4 \rightarrow D3+t$	$S_m^1 \rightarrow \mathbb{R}_t$ (unroll)	I	$\tilde{T}m = 1$	spectral signature
$D3 \rightarrow D2$	forget space	II	area A	—
$D2 \rightarrow D1$	forget space	II	length L	—
$D1 \rightarrow D0$	forget space	II	$n \in \mathbb{Z}$	—

$$\underbrace{T^4 \rightarrow \mathbb{R}^3 \times \mathbb{R}_t}_{\text{Type I: embedding, nearly reversible}} \rightarrow \underbrace{\mathbb{R}^2 \rightarrow \mathbb{R} \rightarrow \{\text{pt}\}}_{\text{Type II: projecting, not reversible}}$$

.8 Three Operations at a Glance

[B]

Type	Operation	Character	Information
I	mass $\leftrightarrow$ time ($\tilde{T}m = 1$)	geometric	nearly reversible
II	spatial projection (down)	geometric	lossy
III	$T^4 \leftrightarrow \mathcal{H}$ /Hilbert space (up)	representational	bijjective

The bijectivity of type III is the mirror image of the non-reversibility of type II: geometrically one loses when reducing, representationally one loses nothing when translating.

.9 The Role of the Recursion ξ^n

[K] The recursion ξ^n is *not* a universal compensation for projection losses – its function is narrowly limited. For type I, ξ^n generates the discrete mass spectrum (particle ladder) – that is the compact mode content of S_m^1 , dual as a time-frequency structure. For type II, ξ^n does not act – spatial projection losses are genuine and not touched by the recursion.

.10 Reversibility, Precisely Per Type

[B] The type-I step is invertible in mode content as a duality (Fourier/KK); the winding remains ambiguous – “nearly reversible”. The spatial sub-chain $\mathbb{R}^3 \rightarrow \{\text{pt}\}$ is not invertible: there is no continuous map $\rho : \{\text{pt}\} \rightarrow \mathbb{R}^3$ with $\rho \circ \Pi = \text{id}$ – every π_k has $\ker = \mathbb{R} \neq \{0\}$, the collapsed fibre structure is no longer present in the image.

.11 Why Exactly Four Circles

[B] No theory derives the dimension number of spacetime; FFGFT does *not derive the 4 forward* – it takes $3 + 1$ as empirical input. The question is therefore not “why is the world four-dimensional?” but: *given $3 + 1$ – what is the minimal structure that carries everything?*

[K] Naively one would need *five* building blocks: three for space, one for a mass spectrum, one for time evolution. But the fundamental relation $\tilde{T} \cdot m = 1$ identifies mass reading and time reading of *one* dimension (type I): the fourth circle S_m^1 carries as a KK tower the mass spectrum *and*, decompactified, time. The duality lets one circle do the work of two – the naive 5 collapses to **4**. Fewer than four (T^3 alone): no mass spectrum, no time. More than four (a second dual circle): a second independent mass tower – superfluous, without correlate. “Optimal” means here *minimally sufficient*: the smallest dimension count in which space, full mass spectrum, and time together arise from one geometric object.

.12 In the Compact Picture: the Time Circle Is Chosen by the Measurement

[S] In the compact T^4 all four circles are identical: dimensionless cyclic coordinates, no signature, no space/time distinction. The asymmetry (one time, signature $(3, 1)$) appears only under decompactification (type I): exactly one circle unrolls via its universal cover into

non-compact time, the other three remain compact (discrete spectrum) or project (space). *Which* circle becomes time is no intrinsic geometric fact – the geometry is symmetric. Time is *relational*: the selection is the measurement process itself. The embedded observer reading from inside experiences one direction as duration; this experiencing constitutes the time circle. The symmetry breaking is the measurement process, not the geometry.

.13 The Topological Identification $t \sim t + R_m$ as Physical Observable

[K] The topological identification is not a mere accompanying structure but a physical observable. In the time reading its period is the fundamental revolution time

$$T_{\text{rev}} = \frac{R_m}{c} = \frac{\xi \ell_P}{c} \approx 7.2 \times 10^{-48} \text{ s}, \quad \Delta\phi = 2\pi\xi \approx 8.4 \times 10^{-4} \text{ rad/revolution.}$$

[K] Two testable readings: (1) **Mass reading (mature, quantitative)**. The discrete mode spectrum generates particle masses as torus overtones; from this follow Koide, $g - 2$, and three generations (A100/A110/A120). Compactness is numerically redeemed. (2) **Time reading (falsifiable)**. The periodicity generates a CHSH deviation from the Tsirelson bound of order ξ ; for 73 qubits $\Delta \approx 5.4 \times 10^{-4}$ (resolvable at $\sigma \sim 10^{-4}$). The IBM Heron r2 hardware test (May 2026) confirms Bell violation ($S = 2.74$); the ξ effect is below the NISQ noise floor and not yet resolvable (A160). No unfalsifiable accompanying structure.

[S] Scaling: linear vs. logarithmic. A naive *coherent linear accumulation* $r \cdot \Delta\phi$ over r circuits would give physically absurd phase drifts ($\sim 10^{44}$ rad/s) and would contradict functioning atomic clocks and deep circuits. The laboratory prediction scales *logarithmically*,

$$\delta\phi \sim \frac{\xi \ln(r)}{D_f} \approx 2 \times 10^{-4} \quad (D_f = 3 - \xi),$$

not linearly. The linear form is the worst-case bound of the RSA argument (coherence over $r \approx N$ cycles), not the laboratory prediction.

.14 The Standard Model as Decompactified Projection

[K] Every computation using α , ν , or Yukawa couplings carries the K_{frak} recursion *indirectly* from the FFGFT perspective: $\alpha = \xi E_0^2$ (A130), where E_0 already absorbs the K_{frak} factor. The masses are the ξ^p -ladder ($m = r \xi^p \nu$, A100); a Yukawa coupling $g = m/\nu$ is already a ξ -ladder ratio. The Standard Model takes these quantities as measured inputs; FFGFT *derives* them. Working with α , ν , or Yukawa in continuous fields on $\mathbb{R}$ is structurally working in the *unrolled* picture – the type-I region of decompactification. The compact origin of the constants is carried along, but not named.

[S] What is established, what remains programme. Established is the containment relation for the sector FFGFT actually derives: α , the lepton masses, the Koide scalar $Q = 2/3$, the phase $\theta = 2/9$ (A120/A130) – for these it is shown that they are recursion outputs. Programme remains the *full* Yukawa matrix: quark masses, mixing angles (CKM/PMNS), CP phase. This separation is the same as throughout the corpus: core derivations (proved from ξ), bridges (algebraically proved), reductions (plausibility sketches).

.15 Open Questions

First, the large-scale outer scale is treated here as a declared external input, not derived from ξ . The two-type structure depends only on the radius-class separation large/small,

not on the numerical value. The connection of the large-scale outer scale to ℓ_1 remains open.

Second, holographic completeness for type II ($D_3 \rightarrow D_2$) is not worked out. To what extent can the type-II loss be reconstructed exactly (not just partially) via boundary data?

Third, the full Yukawa matrix remains programme. The SM-subset relation is a core derivation for the lepton sector; for quark masses/CKM/PMNS/CP-phase it is an open obligation.

.16 Verification Scripts

- Type-I reversibility: Fourier bijection $S_m^1 \leftrightarrow \mathbb{R}_t$ numerically, and winding ambiguity: `python/A_Serie_Skripte/a090_projektion.py`
- CHSH deviation: $\Delta\text{CHSH} \approx \xi/(2\pi)$ numerically and logarithmic vs. linear regime scaling: `python/A_Serie_Skripte/a090_chsh.py`

.17 Supersedes

This document subsumes: Dok. 270 (projection chain $T^4 \rightarrow T^0$, type I/II/III, reversibility, role of ξ^n , why four circles, time circle as measurement process, $t \sim t + R_m$ as observable, linear/logarithmic scaling), Dok. 298 (SM as decompactified projection, Copernicus analogy, what is established/programme). *Not* taken over from Dok. 263 (fractal holography): the bulk with 31 cosmology hits, notably H_0 as projection artefact, vacuum energy, UIFT bridge – all do not belong in the A-series core (A010). Likewise not taken over from Dok. 298: the HLV-YUKAWA section (IPI framework). The holographic projection $D_3 \rightarrow D_2$ (Dok. 263/268/254) is named but not carried out (open point).

.18 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.